

Comparative Analysis of Non-Causal Masked and Autoregressive Large Language Models for Genomic Tasks

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Abstract

Large language models have significantly advanced bioinformatics, with non-causal masked language models excelling in tasks such as genomic sequence alignment, protein structure prediction, and functional genomics by capturing intricate patterns within biological data. These advancements have led to breakthroughs in understanding complex biological systems. Meanwhile, autoregressive language models have demonstrated exceptional performance in general natural language tasks, including text generation, translation, and summarization. Despite their successes, no study has directly compared the performance of non-causal masked language models and autoregressive language models in genomic tasks. This paper introduces GenomeLlama2, an autoregressive language model based on the Llama2 architecture, and trained on genomic data for comparison against foundational non-causal masked language model. We control for parameter counts, context lengths, and other hyperparameters to specifically compare on the attention mask itself on both perplexity and downstream tasks. Evaluated through the comprehensive Genomic Understanding Evaluation (GUE) benchmark, we find that in general non-causal masking outperforms causal/autoregressive masking at the cost of being harder to train. Code and pointers to public data will be available upon publication.

evolutionary biology, and biotechnology. However, the sheer volume and complexity of genomic data present significant challenges, necessitating sophisticated computational models.

Transformers (Vaswani et al., 2023) have revolutionized fields like natural language processing (NLP) and bioinformatics by efficiently managing large datasets and capturing dependencies over both short and long ranges. In genomics, transformer-based non-causal models such as DNABERT2 Zhou et al. (2024) and the Nucleotide Transformer (Dalla-Torre et al., 2023) have significantly advanced the field, providing powerful tools for tasks such as promoter prediction, multi-species genome analysis, and epigenetic marks prediction.

DNABERT-2 (Zhou et al., 2024) is an advanced genomic foundation model that overcomes the limitations of its predecessor, DNABERT (Ji et al., 2021), by replacing k-mer tokenization with SentencePiece (Kudo and Richardson, 2018) with Byte Pair Encoding (BPE) (Sennrich et al., 2016) and introducing architectural enhancements such as flash attention (Dao et al., 2022) and attention mechanisms with linear biases (Press et al., 2022), all of which significantly boost computational efficiency. DNABERT-2 matches state-of-the-art performance with lower computational costs, excelling in genome analysis and handling long DNA sequences.

On the other hand, the Nucleotide Transformer (Dalla-Torre et al., 2023) is a robust foundation model for genomics that leverages transformer-based architectures to encode nucleotide sequences into rich, context-aware representations using DNA bases as tokens. It either matches or surpasses baseline performance in various genomic prediction tasks, including epigenetic marks prediction and variant impact assessment, showcasing its versatility and potential for generalization across different biological contexts.

1. Introduction

The field of genomics has undergone a paradigm shift with the advent of machine learning, enabling unprecedented insights into complex genetic data. Understanding the intricate patterns within genomic sequences is crucial for advancements in medicine,

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Together, DNABERT2 and the Nucleotide Transformer highlight the transformative potential of advanced transformer models in genomics, enabling more accurate and insightful analyses.

Autoregressive models, known for their prowess in NLP tasks, present a promising avenue for genomic analysis. Their ability to generate coherent sequences and model dependencies suggests potential applications in genomics, where sequence generation and prediction are paramount. Recent breakthroughs with models like GPT-3 (Brown et al., 2020) and GPT-4 (OpenAI et al., 2024) have showcased remarkable capabilities in language understanding, generation, and even reasoning, driven by their large-scale training on diverse datasets. The Llama models (Touvron et al., 2023a,b; Dubey et al., 2024), particularly Llama 2 (Touvron et al., 2023b), have also demonstrated exceptional performance by refining transformer architectures to optimize efficiency and scalability, making them well-suited for tasks requiring long-context processing. These advancements highlight the potential for autoregressive models to be adapted for genomics, where their strengths in handling complex sequence data can be leveraged for tasks like variant prediction, functional annotation, and synthetic sequence generation.

This study introduces GenomeLlama2, a transformer-based autoregressive model specifically trained on genomic data. Built on the Llama2 architecture (Touvron et al., 2023b), GenomeLlama2 is tailored to align with DNABERT2 (Zhou et al., 2024) in terms of parameter size and context length, enabling a direct “apples-to-apples” comparison of their performance on genome-related tasks. By leveraging the architectural strengths of Llama2, GenomeLlama2 aims to provide a robust framework for genomic tasks. The model uses the same preprocessed genomic datasets and tokenizer vocabulary as DNABERT2, ensuring consistent and fair comparisons between the two models.

Our research focuses on developing and evaluating GenomeLlama2, comparing its performance with DNABERT2 (Zhou et al., 2024) across various genomic tasks. Furthermore, we explore the impact of varying the parameter size and context length of GenomeLlama2, by training additional models with increased sizes and context lengths, to evaluate their performance relative to DNABERT2.

The study employs the comprehensive Genomic Understanding Evaluation (GUE) benchmark (Zhou et al., 2024) to assess the capabilities of both au-

toressive and non-causal masked language models. This research aims to illuminate the strengths and limitations of these two model types in genomic applications, offering critical insights into which model excels in handling complex genomic data and advancing the field of genomic research. By exploring these differences, the study seeks to contribute to a deeper understanding on how autoregressive and non-causal masked language models perform in genomic applications, revealing their respective advantages and shortcomings.

2. Methods

In this section, we describe the methodologies employed in the development and evaluation of the GenomeLlama2 model, focusing on tokenizer alignment, architectural design, and pretraining processes. By detailing these components, we aim to provide a clear understanding of the steps taken to ensure a fair comparison between GenomeLlama2 and DNABERT2 (Zhou et al., 2024) in genome-related tasks.

2.1. Dataset

GenomeLlama2 is trained using the same dataset that DNABERT2 (Zhou et al., 2024) employs for model pretraining, allowing for a direct comparison between the two models on genome-related tasks. The dataset, as detailed in the DNABERT2 paper, consists of two primary components: the Human Genome Dataset and the Multi-Species Genome Dataset.

The Human Genome Dataset, originally compiled for DNABERT (Ji et al., 2020), serves as a key resource for both models. It is derived from the GRCh38.p13 primary assembly of the reference human genome, sourced from GENCODE (Harrow et al., 2012). Complementing this human-centric dataset, the Multi-Species Genome Dataset introduces a broader genomic perspective by incorporating genomes from 135 species across six distinct categories.

Pretraining GenomeLlama2 on this extensive and diverse dataset offers several benefits. First, the vast array of genomic sequences helps the model recognize complex patterns and structural elements across different species. Additionally, the inclusion of both human and multi-species data enhances the model’s ability to identify conserved genomic features and

species-specific variations, which are crucial for comparative genomics and evolutionary research.

The shared dataset between GenomeLlama2 and DNABERT2 (Zhou et al., 2024) creates a consistent basis for benchmarking and comparative analysis. By using the same training data, researchers can more directly attribute performance differences to the architectural and algorithmic distinctions between the models, rather than to differences in data.

The dataset required no additional preprocessing, as it had already been meticulously prepared. This involved removing sequences containing ambiguous nucleotide symbols, such as “N,” and retaining only the four canonical nucleotides: adenine (A), thymine (T), cytosine (C), and guanine (G). This careful curation ensures the dataset is clean and focused, optimizing the models’ ability to learn and generalize effectively.

2.2. Tokenizer

To ensure a fair and consistent comparison between GenomeLlama2 and DNABERT2 (Zhou et al., 2024), we employed a similar vocabulary for tokenizing the genomic dataset. The tokenizer for GenomeLlama2 was initialized using the same vocabulary set as DNABERT2. This approach was chosen to eliminate any discrepancies in performance due to differences in tokenization strategies, allowing us to focus purely on the architectural contrasts between the two models.

GenomeLlama2 utilizes the SentencePiece (Kudo and Richardson, 2018) with Byte Pair Encoding (BPE) (Sennrich et al., 2016) method for tokenization, which is adopted from the Llama2 architecture (Touvron et al., 2023b). This choice facilitates efficient handling of genomic data, as SentencePiece is well-suited for genomic sequences and can efficiently manage the vast variety of subword units that naturally occur in genetic code. The BPE method segments sequences into subword units, allowing for both common and rare sequences to be effectively captured in the model’s vocabulary.

The DNABERT2 tokenizer, which also uses the SentencePiece (Kudo and Richardson, 2018) with Byte Pair Encoding (BPE) (Sennrich et al., 2016) method for tokenization, is constructed with a vocabulary size of 4,096 tokens. When adapting this vocabulary for GenomeLlama2, the “[MASK]” token was deliberately excluded. The exclusion is due to the fact that GenomeLlama2 is designed as an autore-

gressive model, which does not utilize masked tokens in its training and inference processes.

Consequently, the GenomeLlama2 tokenizer comprises a total of 4,095 tokens. This adjustment was necessary to align the vocabulary with the autoregressive nature of GenomeLlama2 while maintaining as much parity as possible with DNABERT2’s vocabulary. By leveraging the same foundational vocabulary—minus the mask token—we ensure that both models are evaluated on a level playing field with regard to tokenization, enabling a direct comparison of their capabilities in genome-related tasks.

2.3. Architecture

The architecture of GenomeLlama2 is thoughtfully crafted to parallel the capabilities of DNABERT2 (Zhou et al., 2024) while leveraging the strengths of the Llama2 architecture (Touvron et al., 2023b). The objective is to provide a robust model for genome-related tasks, enabling a fair and insightful comparison between autoregressive and non-causal language models. Below, we delve into the architecture of GenomeLlama2.

2.3.1. MODEL CONFIGURATION

GenomeLlama2 builds upon the Llama2 model architecture (Touvron et al., 2023b), with key updates made to some of its parameters. To assess its performance across various genome-related tasks, we trained five variants of GenomeLlama2. The first variant, GenomeLlama2-base, consists of 119 million parameters, closely aligning with the 117 million parameters of DNABERT2, and shares the same context length of 128. This allows for a direct “apples-to-apples” comparison that emphasizes architectural differences rather than disparities in model capacity.

The next two variants, GenomeLlama2-medium and GenomeLlama2-large, are scaled-up versions of GenomeLlama2-base, featuring 411 million and 744 million parameters, respectively, while maintaining the same context length of 128. These models are trained from scratch with their respective parameter counts. This approach allows us to explore how performance scales with compute for GenomeLlama2, and how the models compare with DNABERT2 (Zhou et al., 2024) when all else remain constant.

The final two variants, GenomeLlama2-base-256 and GenomeLlama2-base-512, are derived by extending the context length of GenomeLlama2-base to 256 and 512, respectively. These models are trained

from scratch with their respective context lengths. This approach allows us to investigate the effect of increased context size on the performance of GenomeLlama2-base and its comparative performance against DNABERT2, while keeping other factors unchanged.

Table 1 provides details on the specific parameter updates made to Llama2 for each model variant. For a visual comparison of the number of parameters across the GenomeLlama2 variants, refer to Figure 1.

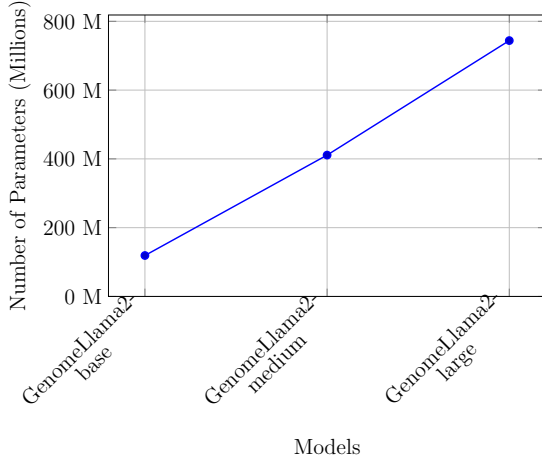


Figure 1: Comparison of parameter counts across different GenomeLlama2 variants

2.3.2. PRE-TRAINING

All variants of GenomeLlama2 are pre-trained using PyTorch Lightning (Falcon and The PyTorch Lightning team, 2019) with multi-GPU support, employing a Distributed Data Parallel strategy (Li et al., 2020) across eight A100 40GB GPUs. These variants are pretrained using the same hyperparameters, with the exception of the maximum sequence length for GenomeLlama2-base-256 and GenomeLlama2-base-512, which are set to 256 and 512, respectively. The other three models, GenomeLlama2-base, GenomeLlama2-medium, and GenomeLlama2-large, have a maximum sequence length of 128, identical to DNABERT2. All models are trained with an effective batch size of 4096.

Like DNABERT2 (Zhou et al., 2024), the training process consists of 500,000 steps and utilizes the AdamW optimizer (Loshchilov and Hutter, 2019), with parameters $\beta_1 = 0.9$, $\beta_2 = 0.98$, an epsilon of

$1e-6$, and a weight decay of $1e-5$. The learning rate is managed by the PyTorch OneCycleLR scheduler (Smith and Topin, 2018), following a linear annealing strategy. It increases linearly from 0 to a peak of $5e-4$ over a warmup period of 30,000 steps and then decreases linearly back to 0 over the remaining 470,000 steps.

2.4. Evaluation

To evaluate GenomeLlama2’s performance, we employed the Genomic Understanding Evaluation (GUE) benchmark (Zhou et al., 2024), a diverse suite of tasks designed to assess models across various genomic challenges. While no single benchmark can guarantee a completely fair comparison, GUE is currently one of the most comprehensive genomic benchmarks available. Our choice of GUE allows for a meaningful comparison with DNABERT2 (Zhou et al., 2024), which has also been evaluated using this benchmark. We acknowledge that a more extensive evaluation across additional datasets and tasks would provide an even more robust comparison, and we consider this as a potential direction for future work.

The GUE benchmark (Zhou et al., 2024) is comprised of a diverse set of tasks that span multiple genomic domains, including human, virus, fungi, yeast, and mouse genomes. These tasks are critical for understanding the functional and structural aspects of genomes and include core promoter detection, transcription factor binding site prediction, splice site prediction, enhance promoter interaction, species classification, transcription factor binding site prediction, epigenetic marks prediction, and covid variant prediction. The details of these evaluation tasks are thoroughly discussed in their paper.

Each task in the GUE benchmark (Zhou et al., 2024) is evaluated using specific metrics to ensure an accurate assessment of model performance. The primary metric used for most evaluation tasks is the Matthews Correlation Coefficient (MCC). However, for the COVID-19 variant classification task, the F1 score is employed.

By utilizing the GUE benchmark (Zhou et al., 2024), we ensure that GenomeLlama2 is rigorously evaluated across a range of tasks, allowing us to draw meaningful comparisons with DNABERT2 (Zhou et al., 2024) and highlight the strengths and potential areas for improvement of autoregressive models like GenomeLlama2 in genomic tasks.

Table 1: Parameter Details of GenomeLlama2 Variants

Model	Hidden Layers	Attention Heads	Key-Value Heads	Hidden Size	Intermediate Size	Context Length
GenomeLlama2-base	12	12	12	768	3072	128
GenomeLlama2-base-256	12	12	12	768	3072	256
GenomeLlama2-base-512	12	12	12	768	3072	512
GenomeLlama2-medium	24	16	16	1024	4096	128
GenomeLlama2-large	28	20	20	1280	5120	128

Each task in the GUE benchmark (Zhou et al., 2024) provides specific training data, enabling the model to be fine-tuned for optimal evaluation on these tasks. All variants of GenomeLlama2 undergo fine-tuning for each task before evaluation. Most hyperparameters from the pretraining stage are retained during fine-tuning, with the exceptions of batch size and learning rate, which align with those used for DNABERT2 (Zhou et al., 2024).

During fine-tuning, a batch size of 32 is used, and the learning rate is managed in the same way as in the pretraining stage, using PyTorch’s OneCycleLR scheduler (Smith and Topin, 2018) with a linear annealing strategy. It starts at 0, increases linearly to a peak of $3e-5$ over a warmup period of 50 steps, and then decreases linearly back to 0 over the remaining steps. The exact number of training steps for each task in the GUE benchmark is detailed in the DNABERT2 paper.

3. Results

In a direct comparison between DNABERT2 (Zhou et al., 2024) and GenomeLlama2-base, where both models are similar in size, DNABERT2 significantly outperforms GenomeLlama2-base on nearly all tasks within the Genomic Understanding Evaluation (GUE) benchmark (Zhou et al., 2024). The only exception is the prediction of the H3K4me2 epigenetic mark, where GenomeLlama2-base achieves better results, as shown in Table 2.

Increasing the size of GenomeLlama2-base up to six times its original size leads to improved overall performance on the GUE benchmark tasks. However, even with this increased model size, GenomeLlama2-base surpasses DNABERT2 only in the prediction of the H3K4me2 epigenetic mark, as presented in Table 2.

Similarly, extending the context length of GenomeLlama2-base to up to four times its original

length further enhances its overall performance on the GUE benchmark. Despite these enhancements, GenomeLlama2-base with increased context length exceeds DNABERT2’s performance only in the H3K4me2 epigenetic mark prediction task, as shown in Table 6.

4. Conclusion

The results of this study demonstrate that non-causal masked language models, such as DNABERT2 (Zhou et al., 2024), are generally more effective than autoregressive models like GenomeLlama2 for genomic tasks. In our experiments, DNABERT2 consistently outperformed GenomeLlama2 across nearly all tasks within the Genomic Understanding Evaluation (GUE) benchmark (Zhou et al., 2024), showcasing its ability to handle complex genomic data with greater accuracy and efficiency. The sole task where GenomeLlama2 excelled was the prediction of the H3K4me2 epigenetic mark, indicating that while autoregressive models may have niche applications, they lack the broader applicability and robustness in the genomic domain.

Moreover, attempts to enhance the performance of GenomeLlama2 by increasing its model size and context length yielded only marginal improvements, further underscoring the inherent advantages of non-causal masked models. These advantages likely stem from their capacity to capture intricate dependencies and contextual relationships within sequences without the constraints of sequential data processing.

In conclusion, the findings suggest that non-causal masked models are better suited for genomic applications, offering superior performance across a wide range of tasks. This research highlights the importance of selecting the appropriate model architecture for specific tasks and provides valuable insights for future developments in the field of genomics.

Table 2: Comparative Analysis of Epigenetic Marks Prediction between DNABERT2 and GenomeLlama2 Variants with Scaled Model Sizes

Model	Epigenetic Marks Prediction						
	H3	H3K14ac	H3K36me3	H3K4me1	H3K4me2	H3K4me3	H3K79me3
DNABERT2	78.27	52.57	56.88	50.52	31.13	36.27	67.39
GenomeLlama2-base	71.14	47.23	51.14	42.03	33.53	33.57	60.58
GenomeLlama2-medium	74.09	48.46	50.79	41.29	31.05	35.80	59.75
GenomeLlama2-large	74.48	49.18	49.37	40.17	34.67	36.14	61.21

Table 3: Comparative Analysis of Epigenetic Marks Prediction, Promoter Detection and Covid Variant Classification between DNABERT2 and GenomeLlama2 Variants with Scaled Model Sizes

Model	Epigenetic Marks Prediction			Promoter Detection			Virus
	H3K9ac	H4	H4ac	all	nontata	tata	covid
DNABERT2	55.63	80.71	50.43	86.77	94.27	71.59	71.02
GenomeLlama2-base	52.79	79.87	43.67	85.45	92.65	61.87	56.92
GenomeLlama2-medium	51.59	79.01	45.85	85.44	93.48	63.56	57.12
GenomeLlama2-large	53.58	78.88	44.51	85.22	94.19	60.53	57.48

Table 4: Comparative Analysis of Transcription Factor Prediction for Human and Core Promoter Detection between DNABERT2 and GenomeLlama2 Variants with Scaled Model Sizes

Model	Transcription Factor Prediction (Human)					Core Promoter Detection		
	0	1	2	3	4	all	nontata	tata
DNABERT2	71.99	76.06	66.52	58.54	77.43	69.37	68.04	74.17
GenomeLlama2-base	66.55	65.15	57.40	52.62	68.40	60.25	65.97	43.96
GenomeLlama2-medium	65.11	67.20	59.82	50.80	70.45	59.22	65.62	54.01
GenomeLlama2-large	64.01	66.57	54.40	47.23	68.86	60.50	66.46	45.88

Table 5: Comparative Analysis of Transcription Factor Prediction for Mouse and Splice Site Detection between DNABERT2 and GenomeLlama2 Variants with Scaled Model Sizes

Model	Transcription Factor Prediction (Mouse)					Splice
	0	1	2	3	4	Reconstruct
DNABERT2	56.76	84.77	79.32	66.47	52.66	84.99
GenomeLlama2-base	51.18	81.31	71.34	65.70	43.82	77.44
GenomeLlama2-medium	53.08	84.26	74.41	50.90	43.30	83.41
GenomeLlama2-large	55.33	84.53	73.21	42.65	48.28	81.50

Table 6: Comparative Analysis of Epigenetic Marks Prediction between DNABERT2 and GenomeLlama2 Variants with Scaled Context Length

Model	Epigenetic Marks Prediction						
	H3	H3K14ac	H3K36me3	H3K4me1	H3K4me2	H3K4me3	H3K79me3
DNABERT2	78.27	52.57	56.88	50.52	31.13	36.27	67.39
GenomeLlama2-base	71.14	47.23	51.14	42.03	33.53	33.57	60.58
GenomeLlama2-base-256	73.54	46.66	52.42	43.93	33.03	35.49	62.77
GenomeLlama2-base-512	73.41	47.05	49.25	41.39	33.42	31.94	60.65

Table 7: Comparative Analysis of Epigenetic Marks Prediction, Promoter Detection and Covid Variant Classification between DNABERT2 and GenomeLlama2 Variants with Scaled Context Length

Model	Epigenetic Marks Prediction			Promoter Detection			Virus
	H3K9ac	H4	H4ac	all	nontata	tata	covid
DNABERT2	55.63	80.71	50.43	86.77	94.27	71.59	71.02
GenomeLlama2-base	52.79	79.87	43.67	85.45	92.65	61.87	56.92
GenomeLlama2-base-256	52.66	79.06	44.97	85.97	93.74	59.17	70.59
GenomeLlama2-base-512	51.40	79.73	40.63	85.88	93.67	61.47	70.92

Table 8: Comparative Analysis of Transcription Factor Prediction for Human and Core Promoter Detection between DNABERT2 and GenomeLlama2 Variants with Scaled Context Length

Model	Transcription Factor Prediction (Human)					Core Promoter Detection		
	0	1	2	3	4	all	nontata	tata
DNABERT2	71.99	76.06	66.52	58.54	77.43	69.37	68.04	74.17
GenomeLlama2-base	66.55	65.15	57.40	52.62	68.40	60.25	65.97	43.96
GenomeLlama2-base-256	66.03	71.40	60.40	50.28	68.64	59.87	63.87	44.95
GenomeLlama2-base-512	66.83	71.18	59.76	48.54	71.01	58.91	64.73	44.10

Table 9: Comparative Analysis of Transcription Factor Prediction for Mouse and Splice Site Detection between DNABERT2 and GenomeLlama2 Variants with Scaled Context Length

Model	Transcription Factor Prediction (Mouse)					Splice
	0	1	2	3	4	Reconstruct
DNABERT2	56.76	84.77	79.32	66.47	52.66	84.99
GenomeLlama2-base	51.18	81.31	71.34	65.70	43.82	77.44
GenomeLlama2-base-256	54.58	82.12	76.92	59.95	44.68	82.56
GenomeLlama2-base-512	54.59	82.80	68.33	57.75	45.61	82.71

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